light, and $M_\odot = 1.989 \times 10^{30}$ kilograms is the mass of the Sun. See, e.g., Chapters 31 and 32 of MTW.

35 [In honor of those tides, . . . *tidal force.]* The tidal force, expressed as a relative acceleration between your head and feet (or between any other two objects), is $\Delta a = 16\pi^3 G(M_h/C^3)L$, where G is Newton's gravitation constant (see above), M_h is the black-hole mass, C is the circumference at which you are located, and L is the distance between your head and feet. Note that 1 Earth gravity is 9.81 meters per second^2. See, e.g., page 29 of MTW.

37 [General relativity predicts, . . . actually decreases.] The above formula (note to page 35) gives for the tidal force $\Delta a \propto M_h/C^3$. When the circumference is nearly that of the horizon, $C \propto M_h$ (note to page 28), so $\Delta a \propto 1/M_h^2$.

37 [The entire trip of 30, 100 light-years . . . only 20 years.] Starship time T_{ship}, Earth time T_E, and distance D traveled are related by $T_E = (2c/g)\sinh(gT_{ship}/2c)$ and $D = (2c^2/g)[\cosh(gT_{ship}/2c) - 1]$, where g is the ship's acceleration ("one Earth gravity," 9.81 meters per second^2), c is the speed of light, and cosh and sinh are the hyperbolic cosine and hyperbolic sine functions. See, e.g., Chapter 6 of MTW. For trips that last much more than one year, these formulas become, approximately, $T_E = D/c$ and $T_{ship} = (2c/g)\ln(gD/c^2)$, where In is the natural logarithm.

39 [To remain in a circular orbit, . . . hurled you inward.] For a mathematical analysis of circular (and other) orbits around a nonspinning black hole, see, e.g., Chapter 25 of MTW, and especially Box 25.6.

40 [Your calculations show . . . 1.0001 horizon circumferences.] The acceleration force you will feel, hovering at a circumference C above a black hole of mass M_h and horizon circumference C_h, is $a = 4\pi^2 G(M_h/0) \times (1/\sqrt{1 - C_h/C})$, where G is Newton's gravitation constant. If you are very close to the horizon, then $C \simeq C_h \propto M_h$, which implies $a \propto 1/M_h$.

40 [Using the usual 1-g acceleration . . . crew in the starship.] See the second note for page 37 above.

43 [The spot is small . . . seen from Earth.] When one hovers at a circumference C slightly above a horizon with circumference C_h one sees all the light from the external Universe concentrated in a bright disk with angular diameter $\alpha \simeq 3\sqrt{3}\sqrt{1 - C_h/C}$ radians $\simeq 300\sqrt{1 - C_h/C}$ degrees. See, e.g., Box 25.7 of MTW.

44 [Equally peculiar, the colors . . . 5×10^{-7} meter light.] When one hovers at a circumference C slightly above a horizon with circumference C_h, one sees the wavelengths l of all light from the external Universe gravitationally blue-shifted (the inverse of the gravitational redshift) by $\lambda_{received}/\lambda_{emitted} = \sqrt{1 - C_h/C}$. See, e.g., page 657 of MTW.

49 [Inserting these numbers . . . coalesce seven days from now.] When two black holes, each with mass M_h, orbit each other with separation D, they have an orbital period $2\pi\sqrt{D^3/2GM_h}$, and their gravitational-wave recoil forces them to spiral together and